

FAD: A One-Stop Database for Food Allergen Research and Information

Wafa Mokhtari^{(1,2)*},
Souad Khemili-Talbi⁽³⁾,
Jean Kwasigroch⁽¹⁾,
Dimitri Gilis⁽¹⁾.

Belgium, Algeria

^{*(1)}Computational Biology and Bioinformatics, Université libre de Bruxelles, Belgium.

^{*(2)} Biological Resources Valorization and Conservation Laboratory “VALCORE”, M’Hamed Bougara University of Bouverdes, Algeria.⁽³⁾ Laboratoire de Bioinformatique, Microbiologie Appliquée et Biomolécules (BMAB), Algeria
Email: wafa.mokhtari@ulb.be, w.mokhtar@univ-bouverdes.dz



Summary:

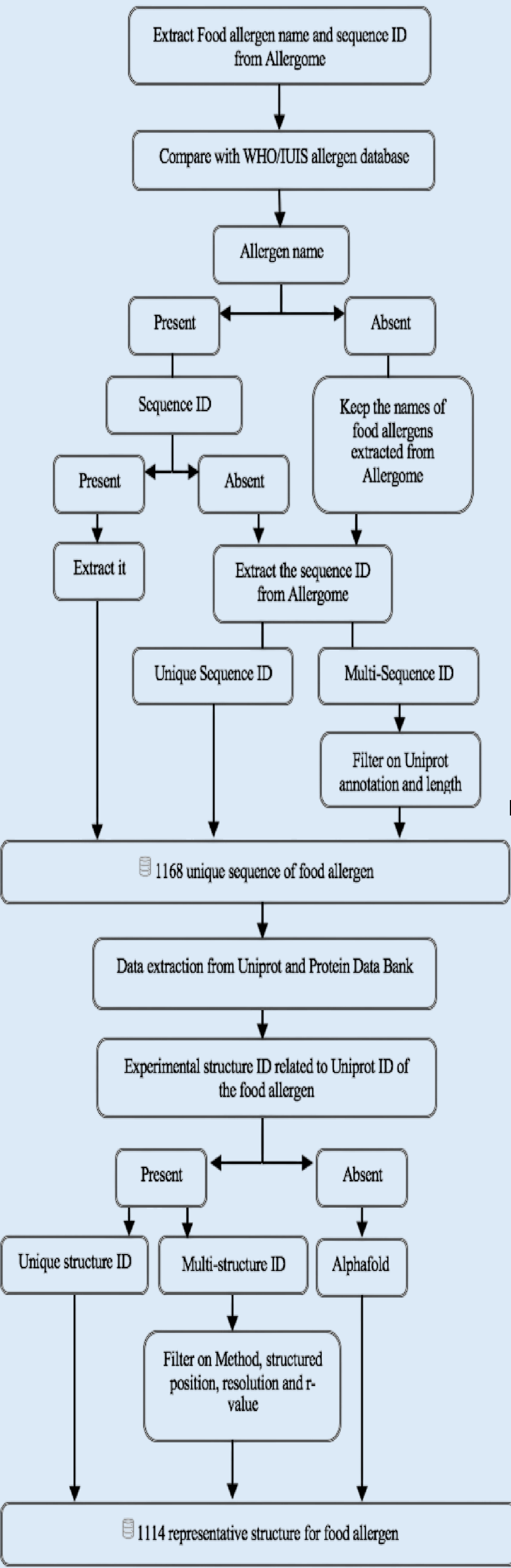
Food allergies impact approximately 220 million individuals worldwide[1], underscoring the necessity for a centralized allergen database. The Food Allergens Database (FAD) addresses this gap by consolidating essential information, including 1,168 unique allergen sequences and 2,456 epitopes, systematically extracted from reputable sources. FAD enhances accessibility for researchers and developers by facilitating customized data downloads. This comprehensive resource significantly advances food allergen research and safety, serving as an indispensable tool for scientists and healthcare professionals, with future expansions aimed at incorporating additional data and features.

Introduction: Food allergies are a growing public health issue, affecting millions of people worldwide. Allergic reactions can range from mild to potentially life-threatening, making the availability of a specialized food allergen database essential[2]. The problem statement is the lack of a dedicated food allergen database and the need to consult multiple databases to obtain comprehensive information.

Goals:

To create a database that integrates the biochemical, structural, immunological, and functional properties of food allergens. To facilitate access to comprehensive and up-to-date information on food allergens with customized download options.

Methods/Approach



Flowchart depicting the development of the Food Allergen Database.



Scan Me !

References

[2]Fiocchi A, Sampson HA et al. “Food Allergy”, Section 2.5 in WAO White Book on Allergy, Editors: R Pawankar, GW Canonica, ST Holgate, RF Lockey (Milwaukee, Wisconsin: World Allergy Organization, 2011), pp 47-53.
[2] Boughellout, H., et al.
"Prévalence de l'allergie alimentaire chez une population âgée de 0 à 15 ans de la ville de Skikda Algérie." *Revue Française d'Allergologie* 63.3 (2023): 103357.

Results and Impacts

1.Centralized Information : The FAD offers centralized access to 1,168 allergen sequences, 1,114 structures, and 2,456 epitopes, enabling fast and reliable data retrieval.

2.Comprehensive Integration : The database integrates biochemical, structural, immunological, and functional properties of food allergens for a complete understanding.

3.Boosts Research : It supports scientists by offering resources to study food allergens, develop innovative treatments or diagnostics.

FOOD ALLERGEN DATABASE

The **Food Allergen Database** (FAD) is dedicated to providing valuable information on existing food allergens, aiming to assist scientists, regulators, the food industry and allergic consumers. FAD gathers information on the sequence, structure and epitope of food allergens. Refer to the 'Help' menu for further information.

Submit your query: enter a food allergen name (for example, 'Bos d 5') or a Uniprot ID (for example 'P00711'). Type 'Full' to access the entire database or to **download** data.

Submit

GeneralStructureEpitope

General information

Allergen information

Allergen name : Tri a 14
Allergen parent: Tri a 14
Common name : Wheat
Scientific name : Triticum aestivum
The WHO/IUIS Allergen Nomenclature : Recognized
Allergenic tissue(s) : Seed
Enzyme classification : -
Taxonomy ID : 4565

Sequence information

Uniprot ID : [P24296](#)
Allergome ID : [1059](#)
NCBI ID :
Protein name : Non-specific lipid-transfer protein
Uniprot annotation : Reviewed
Sequence lenght : 113 amino acids

InterPro classification

IPR036312 : Bifunctional inhibitor/plant lipid transfer protein/seed storage helical domain superfamily
IPR016140 : Bifunctional inhibitor/plant lipid transfer protein/seed storage helical domain
IPR000528 : Plant non-specific lipid-transfer protein/Par allergen

SupFam sequence superfamily

SSF47699 : Bifunctional inhibitor/lipid-transfer protein/seed storage 2S albumin

FAD home-submission page

FAD General information page

Input type : Uniprot ID, Food allergen name or to the entire database

PDB representative for Tri a 14
PDB code : 1BWO
Method: X-ray
Resolution: 2.10 Angstrom
Chain(s): A/B
Sequence position: 24-113
R-value: 0.213

PDB	Method	Resolution	Chain	Sequence position	R-value
1CZ2	NMR	-	A	24-113	-
1GH1	NMR	-	A	24-113	-
P24296	AlphaFold	-	-	1-113	-
1BWO	X-ray	2.10	A/B	24-113	0.213

FAD Structural information page

Mapping Epitopes of Tri a 14 on the 3D Representative Structure "Biological assembly"

Red: Epitope(s)

Blue: Non-epitope(s)

IEDB ID	Epitope type	Start position	End position	Epitope amino acids	Start mapping	End mapping
190776	Linear	48	57	GQCCDGVKNL	25	34
190905	Linear	60	69	QARSQSDRQS	37	46
117314	Linear	104	113	ISLNDCSRV	81	90

FAD Epitop information page

Please select the food allergen you are interested in from the table containing the entire database below. You may also filter this table with the following four menus. Please note that if multiple filters are selected, the **logical operator OR will be applied**.

Common Name

Enzyme Classification

SupFam

Allergenic Tissues

Submit

Click [this link](#) to download the table below in csv format.

If you want to customize the fields of the table you want to download, click on "Advanced download of the entire database".

Advanced Download of the entire database

If you want to download epitopes of allergens for which they are referenced in IEDB, the PDB file of the representative 3D structure of the allergens, the PDB files of all the experimental 3D structure of the allergens or the PDB files of all the modelled 3D structure of the allergens, click on "Additional Downloads".

Additional Downloads

FAD Full database page

Filterbased on four parameters. Customized food allergen information download.

Conclusion

The FAD database offers a unified and comprehensive repository of critical data. It provides an exhaustive overview with a single platform.

The database’s user-friendly interface facilitates efficient search and download functionalities.

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